

+ SPONGE CITY

01. WHAT I AM ATTEMPTING

I want to ask what happens to an urban surface when it meets a rainstorm, and how that changes depending on the severity of the rainstorm. A grid will track how much rain each cell retains, sends to a catch basin, or passes downslope. Additional variables could include the permeable surface as well as the available drainage in the area.

02. THE DATASET

Spatial joins of public NYC layers made for terrain mapping and stormwater planning. For example:

FIELD	UNIT	SOURCE / DATE
elevation	ft	NYC OTI DEM (2010)
impervious	%	NYC DEP (2020)
catch-basin dist.	ft	NYC DEP (2025)
GI area	ft ²	NYC DEP (monthly)
rain depth	mm	NOAA Atlas
flood-map class	class	NYC DEP (2024)

03. RESEARCH

NYC used a rain-on-mesh model in Phase 1 of its stormwater study to route rain across the surface before adding the sewer network in Phase 2. The City's flood maps provide a comparison layer, not training data.

CADDIES divides terrain into regular square cells and transfers water between neighboring cells according to differences in water level. This is probably the closest precedent for your cell-by-cell update rule and Qin/Qout variables.

CityCAT is a two-dimensional urban flood model that represents buildings, green space, permeable surfaces and flood interventions. It offers a more physically rigorous precedent for testing how changes to the city's surface alter water pathways.

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04. RECORD

Each starting row is one 50 m cell at t = 0.
Fields: cell_id; x/y; elevation_ft;
impervious_pct; catch_basin_dist_ft;
GI_area_ft²; downslope_neighbor;
capacity_remaining_mm; water_depth_mm = 0.

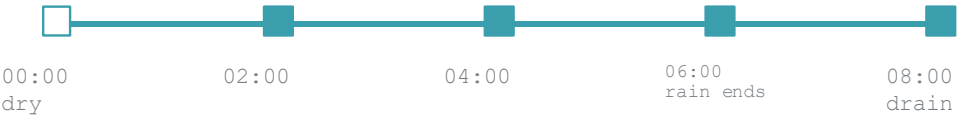
05. RULE

Every five minutes, a cell receives rainfall
and incoming runoff. It removes infiltration
and drainage. Water above local storage is sent
to the lowest of its eight neighbors.

$$S(t+1) = S(t) + R + Q_{in} - I - D - Q_{out}$$

06. RUN

E.g. A six-hour storm runs from 00:00 to 06:00 in
five-minute steps, followed by two hours of
drainage. Compare 2-, 10- and 100-year six-hour
Atlas 14 depths. Rain is distributed evenly over
time to simplify the model



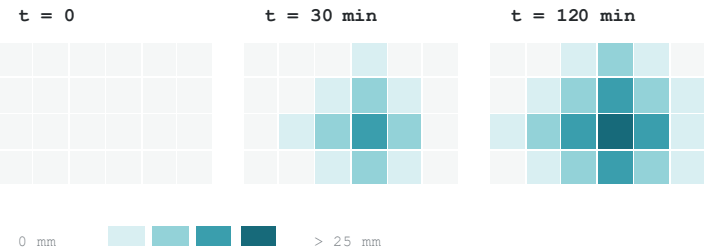
07. THE DATA GENERATED

Illustrative rows only. Each timestep appends one
row per cell.

cell	time	rain	I	D	Q _{in}	Q _{out}	depth
G17	00:05	4.5	0.3	1.2	0.0	1.8	1.2
G17	00:10	4.5	0.3	1.2	2.0	1.8	4.4
G18	00:10	4.5	1.2	0.0	1.8	2.4	2.7
G19	00:10	4.5	2.0	1.0	2.4	0.5	3.4

e.g. new fields: rain_mm; infiltration_mm; drained_mm;
incoming_mm; outgoing_mm; water_depth_mm;
capacity_remaining_mm; receiver_id

07. MAP SEQUENCE



08/WHAT IT CANNOT SEE

- pipe connectivity, capacity or backflow
- clogged inlets, pumps and operations
- soil saturation before the storm
- rain varying across the study area
- tides or coastal backwater
- The output describes water pathways under declared assumptions. It does not establish property-level flood risk or who experiences the result as disaster.

